# Controlled Training-Data Revision: Policy and Grounding Variation

**Status:** Owner built, validated, trained and evaluated; candidate not promoted. Date: 2026-10-16.

## Hypothesis and Scope

The training audit found correct but narrow state-grounding examples and a retained urgency-policy defect. Prepare a separate data-quality experiment before considering more epochs. This is a combined policy/wording/state intervention; an eventual result cannot identify which of those components caused any change.

Only eligible BAYMAX rows inside the existing 1,000-row training cap change. Preserve their positions and broad classes, all REFUSE targets, all xLAM rows, the rest of the training file, and validation/test bytes. No resplitting, new examples from held-out fixtures, oversampling, or new framework is introduced.

The reference is unweighted two-epoch QKVO name-first SFT, config hash `684a76cd39d845b6177bdaee495bb77bcb436406d384baf912f80b8164fb8b8b`. Its parent dataset revision is `685a6b67832a5c55f54f2fc529f5c5234a05baa93d3e0cf241c38c006b9bbc12`. The transform version is `training-grounding-v1`, not a claim that the whole dataset was regenerated by generator 15 or 16.

## Previewed Changes

The model-free preview on the actual capped pool applies 49 changed rows:

| Generic rule | Rows | Change |
|---|---|---|
| Explicit deadline clarification | 3 | Bare "soon" requests become explicit "due soon" requests, so their CLARIFY target is appropriate. |
| Missing-fields wording | 4 | Rephrase calendar requests and questions while retaining both missing time and duration. |
| Undated-task ACT | 10 | Replace eligible dated task requests with explicit undated urgency requests; omit due_date without inventing one. |
| Entity-state variation | 27 | Preserve the uniquely named target, vary its internal ID and position, and add an unrelated entity. Calendar distractors use a different date to avoid inventing a scheduling conflict. |
| Clipboard-content binding | 5 | Vary instruction wording and clipboard values while binding the task title or email body to that state. Task cases also include an unrelated existing task. |

One additional undated-task candidate would duplicate an earlier revised input. Its original dated parent is retained unchanged. The change journal records this skip explicitly; provenance counts distinguish applied changes from skipped rows.

The urgency correction does not relabel the three CLARIFY rows into ACT, which would change quotas. It revises their requests to genuinely require a deadline; separate existing ACT slots teach the agreed undated behavior. This is a new training artifact, not a rewrite of evaluation targets or old results.

Task/clipboard variants use independent stock/equipment text and generic phrasing, not held-out scenario solutions. Unspecified draft subjects are omitted where the subject is not required; supplied recipients and content remain grounded.

No claim of sophisticated semantic diversity is made: the bank remains small, and structural row uniqueness is not a measure of conceptual independence.

## Fixed Controls

| Control | Retained setting |
|---|---|
| Raw cap | 1,000 rows |
| BAYMAX contribution | 400: 200 ACT, 100 CLARIFY, 100 REFUSE |
| External capped contribution | Same 600 xLAM rows in the same slots |
| Base model | Qwen/Qwen2.5-1.5B-Instruct |
| LoRA | r=8, alpha=16, Q/K/V/O, dropout=0.05 |
| Loss weighting | Unweighted, first-name weight 1 |
| Learning rate / epochs / seed | 2e-4 / 2 / 1 |
| Laptop settings for a later run | Context 1,024, batch 1, accumulation 16 |
| Formatting / policy prompt | Same compact JSON, name-first targets and system prompt |
| Validation cap | Same first 200 rows; full validation/test files preserved |

Raw counts are preserved, not promised post-tokenization counts. Changed wording and additional state change token exposure. Inspect the actual tokenized artifact before training; do not assume the old 974 usable rows still apply automatically.

## Preparation and Integrity

`scripts/prepare_sft_grounding.py` uses a focused training-only derivation in `baymax.models.sft_grounding`. It rejects populated destinations, validates the reference identity, copies its training settings into a newly hashed config, and records source/output file hashes plus every changed row's before/after SFT messages. Parent source identities and split membership remain unchanged; the change report records transformed content rather than misrepresenting parent row hashes as new-content hashes.

Changed targets pass the existing strict contract validator and in-memory fake adapter/state validation. No live tool or external API executes. Full preparation rejects revised inputs that duplicate training inputs or overlap any protected validation/test input. It checks source counts, prompt hashes, source stability, and exact held-out byte preservation before publishing from a staging directory. Exact overlap checks do not prove semantic/template separation.

The preview validates the capped changes and counts only. It creates no dataset, config, model or tokenizer and does not replace the full preparation checks. Tiny fixture tests cover deterministic rules, grounded arguments, class/slot preservation, unchanged held-outs, duplicate/overlap rejection, and no-overwrite behavior. No full project build or training was run by the coding agent.

## Owner-Run Next Step

From `project-baymax`:

```
.venv\Scripts\python scripts\prepare_sft_grounding.py

.venv\Scripts\python scripts\validate_sft_dataset.py `
  --dataset-dir data\sft\v1\name-first-grounding-v1
```

Preparation creates these new paths:

- `data/sft/v1/name-first-grounding-v1/`
- `configs/sft/qwen2_5_1_5b_lora_grounding_v1_epochs2.json`
- A future training output named `models/qwen2_5_1_5b_lora_grounding_v1_epochs2/`.

The dataset contains `grounding-changes.json` for before/after review and `grounding-provenance.json` for counts, rules and parent/output identity. Review the generated revision and post-tokenization report before starting training. Do not overwrite old artifacts or treat preview counts as benchmark improvements.

The existing unweighted checkpoint remains the reference. Any later comparison must retain compact inference, generation settings and scoring. The repeatedly inspected 150-case suite is development evidence, not an untouched final test. Local experiment Markdown and generated datasets are not staged or committed.

## Owner-Reported Outcome

The owner validated revision `91ce08228633738365461a556668a6883b5faa806b84d969dddd8164114603a1`, with training config hash `f060ddbc1df6182840086ffc4b60f27785390ee2ee41938e2578bea3f3663986`. The 1,024-token artifact retained 974 training rows and 195 validation rows, including all 100 training clarification and 100 refusal rows, with zero exact duplicates. The owner then completed the two-epoch training and compact evaluation.

| Run | Successes / 150 | Macro-F1 | Invalid / 150 | Correct ACT / 81 | Correct CLARIFY / 40 | Correct REFUSE / 29 |
|---|---|---|---|---|---|---|
| Unweighted reference | 57 | 0.612 | 1 | 66 | 10 | 23 |
| Grounding v1 | 53 | 0.591 | 0 | 66 | 10 | 21 |

Raw artifacts: unweighted and grounding v1. These figures come from the owner's comparison, not an invented estimate or a new comparison run by the coding agent. The revision did not improve the measured result; keep the unweighted model as the development reference. A single seed and four fewer successes do not establish a consistent regression or identify which revision component caused it. The corrections remain in a separate artifact.

The next 2k coverage experiment uses grounding v1 as its corrected 1k parent. Compare against that parent to measure expansion, while also retaining the stronger unweighted reference for context.

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Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.